



DATA QUALITY

A matter of eyes

1

What is DQ

2

DQ dimensions

3

DQ Metrics

4

DQ
Assessment

5

DQ
Improvement

QUESTION I

- What is Data Quality in a nutshell?

ISSUE FROM LOW DATA QUALITY

- The US government found that the third-leading cause of death in their country is patient misidentification. Because the number of duplicate records in hospitals is increasing. This results in the misidentification of 10% of incoming patients, and sharing patient data from disparate sources further increases the probability of duplicate records generation.
- Network for Excellence in Health Innovation, prescription errors resulted in a cost of \$21 billion per year
- The Data Warehousing Institute estimates that bad, inaccurate, dirty or missing data costs companies more than \$600 billion a year

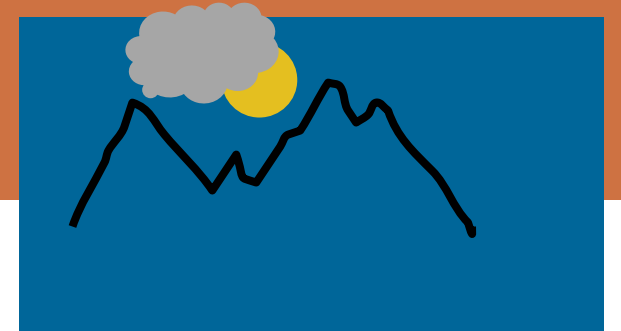
EXAMPLE OF POOR DATA QUALITY

- Every year, poor data quality costs organizations an average \$12.9 million. (gartner 2021)



Want to learn more? Visit www.dataladder.com.

DATA DESCRIBE REAL WORLD



**Linguistic or
sensorial based structure**



Representation model



I saw a marvelous
yellow and red sunset



WHAT IS THE BEST PICTURE OF SARS?



www.hoax-slayer.com



astrobiology.nasa.gov

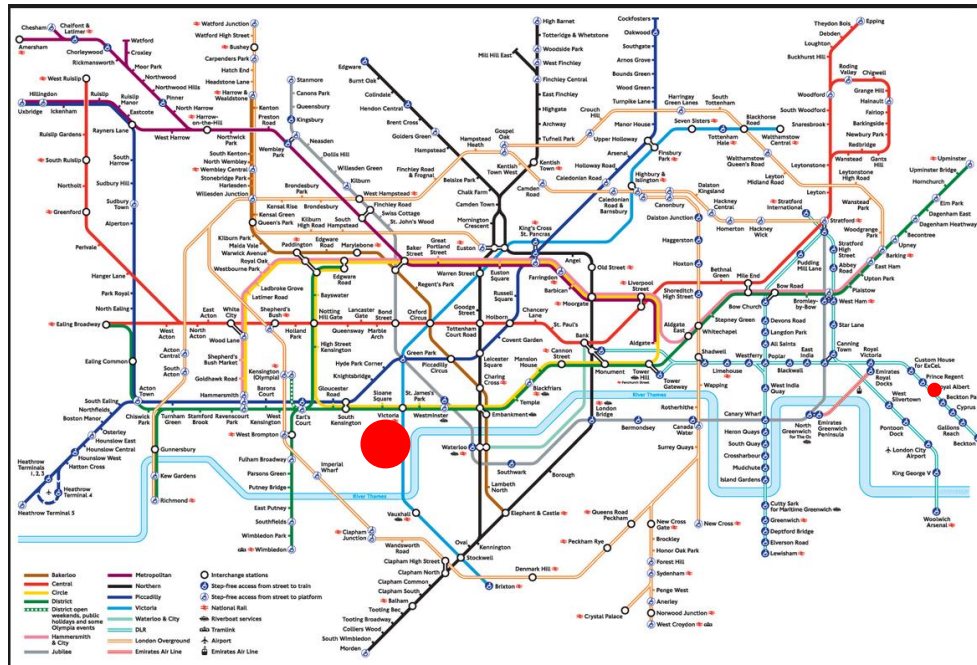
The first image was downloaded from an amateur blog, the second from the NASA site. Your assessments are probably based on your "Mars model".
Now that you know which sites to differentiate from, are you thinking of changing your opinion?

SUBWAY MAPS DI LONDRA E NEW YORK

WHAT IS THE BEST ONE?



IF YOU ARE IN LONDON - DOWNING STREET OR NEW YORK WASHINGTON SQUARE AND YOU WANT TO FIND THE NEAREST STATION....



USEFULNESS VS FAITHFULNESS

Usefulness – precision of the internal representation of the image w.r.t. the task performed.

Faithfulness – adherence to reality



Here the task is to understand the rules of parking
The second image is more useful than the first one

FUNDAMENTAL CONCEPTS

- Quality → characteristics of an artifact that bear on its ability to satisfy stated or implied user needs and expectation
- Dimension (or Characteristic, synonym) → A specific characteristic describing the quality of information, usually not measurable. Sometimes classified in terms of...
- Subdimension (Subcharacteristic)
- Metrics, we refer to the definitions given within the ISO standard 9126-1 and the ISM3 framework
 - a set of elements encompassing both
 - a measurement procedure (or method) i.e., an algorithm that takes the element to measure and associates it with a measure (be it ordinal or interval value), and
 - a proper unit of measure (or scale), i.e., the domain of values returned by the measuring procedure.

In general, several metrics can be associated to each quality dimension



DATA QUALITY DIMENSIONS



FIRST ISSUE - HOW MANY DIMENSIONS HAS DATA QUALITY?

AN EXAMPLE IN THE DOMAIN OF MOVIES, DIRECTORS, REMAKES

Id	Title	Director	Year	#Remakes	LastRemakeYear
1	Casablanca	Weir	1942	3	1940
2	Dead Poets Society	Curtiz	1989	0	NULL
3	Rman Holiday	Wylder	1953	0	NULL
4	Salvina	NULL	1964	0	1985

Inaccurate

Incomplete

Inconsistent

Inconsistent

LOTS OF DIMENSIONS MENTIONED IN METHODOLOGIES..

SEE BATINI ET AL. 2009

Acronym	Data Quality Dimension
TDQM	Accessibility, Appropriateness, Believability, Completeness, Concise/Consistent representation, Ease of manipulation, Value added, Free of error, Interpretability, Objectivity, Relevance, Reputation, Security, Timeliness, Understandability
DWQ	Correctness, Completeness, Minimality, Traceability, Interpretability, Metadata Evolution, Accessibility (System, Transactional, Security), Usefulness (Interpretability), Timeliness (Currency, Volatility), Responsiveness, Completeness, Credibility, Accuracy, Consistency, Interpretability
TQDM	Inherent dimensions: Definition conformance (consistency), Completeness, Business rules conformance, Accuracy (to surrogate source), Accuracy (to reality), Precision, Nonduplication, Equivalence of redundant data, Concurrency of redundant data, Pragmatic dimensions: accessibility, timeliness, contextual clarity, Derivation integrity, Usability, Rightness (fact completeness), cost.
AIMQ	Accessibility, Appropriateness, Believability, Completeness, Concise/Consistent representation, Ease of operation, Freedom from errors, Interpretability, Objectivity, Relevancy, Reputation, Security, Timeliness, Understandability
CIHI	Dimensions: Accuracy, Timeliness Comparability, Usability, Relevance Characteristics: Over-coverage, Under-coverage, Simple/correlated response variance, Reliability, Collection and capture, Unit/Item non response, Edit and imputation, Processing, Estimation, Timeliness, Comprehensiveness, Integration, Standardization, Equivalence, Linkage ability, Product/Historical comparability, Accessibility, Documentation, Interpretability, Adaptability, Value.
DQA	Accessibility, Appropriate amount of data, Believability, Completeness, Freedom from errors, Consistency, Concise Representation, Relevance, Ease of manipulation, Interpretability, Objectivity, Reputation, Security, Timeliness, Understandability, Value added.
IQM	Accessibility, Consistency, Timeliness, Conciseness, Maintainability, Currency, Applicability, Convenience, Speed, Comprehensiveness, Clarity, Accuracy, Traceability, Security, Correctness, Interactivity.
ISTAT	Accuracy, Completeness, Consistency
AMEQ	Consistent representation, Interpretability, Case of understanding, Concise representation, Timeliness, Completeness Value added, Relevance, Appropriateness, Meaningfulness, Lack of confusion, Arrangement, Readable, Reasonability, Precision, Reliability, Freedom from bias, Data Deficiency, Design Deficiency, Operation, Deficiencies, Accuracy, Cost, Objectivity, Believability, Reputation, Accessibility, Correctness, Unambiguity, Consistency
COLDQ	Schema: Clarity of definition, Comprehensiveness, Flexibility, Robustness, Essentialness, Attribute granularity, Precision of domains, Homogeneity, Identifiability, Obtainability, Relevance, Simplicity/Complexity, Semantic consistency, Syntactic consistency. Data: Accuracy, Null Values, Completeness, Consistency, Currency, Timeliness, Agreement of Usage, Stewardship, Ubiquity, Presentation: Appropriateness, Correct Interpretation, Flexibility, Format precision, Portability, Consistency, Use of storage, Information policy: Accessibility, Metadata, Privacy, Security, Redundancy, Cost.
DaQuinCIS	Accuracy, Completeness, Consistency, Currency, Trustworthiness
QAFD	Syntactic/Semantic accuracy, Internal/External consistency, Completeness, Currency, Uniqueness.
CDQ	Schema: Correctness with respect to the model, Correctness with respect to Requirements, Completeness, Pertinence, Readability, Normalization, Data: Syntactic/Semantic Accuracy, Semantic

WE WILL CONSIDER

Cluster of Dimensions	Abstract Definition
Accuracy, Correctness, Precision	Proximity of data in representing a given reference
Completeness, Pertinence	Data represent all (and only) the aspects of the reality of interest.
Minimality, Redundancy, Compactness	Data represent all the aspects of the reality of interest only once and with the minimal use of resources.
Consistency, Coherence	Data comply to all the properties of their membership set (class, category,...) as well as to those of the sets of elements the reality of interest is in some relationship.
Readability, Comprehensibility, Usability	Data are easily perceivable and understood by users.
Accessibility, Usability	Data can be effectively “exploited” (consumed) by users.
Currency, Volatility, Timeliness	Data are up-to-date

SECOND ISSUE: OBJECTIVE VS SUBJECTIVE MEASURES

- Objective measures
 - formal and precise ways to measure metrics for a quality dimension in terms of the values of a domain.
 - independent of human perception/assessment.
- Subjective measures
 - ways to measure metrics for a quality dimension depending on the perception (usually explicit and ordinal rating) of people involved in the measurement process.
 - dependent on human perception/assessment.



DATA QUALITY DIMENSION IN THE RELATIONAL MODEL



CONTENTS

- Data Quality dimensions and metrics
 - Accuracy
 - Completeness
 - Currency
 - Consistency
 - Tradeoffs between dimensions
- Quality assessment and improvement
 - Case study



ACCURACY



ACCURACY OF VALUES

	Accuracy	Completeness	Currency	Consistency
Alphanumeric value	X			
Numerical value				
Tuple	X			
Relation	X			

DEFINITION OF ACCURACY

- Accuracy of a value v is defined as the **closeness** between v and a value v' considered as **the correct representation of the real world phenomenon** that v aims to represent.
- As an example,
 - if the name of a person is “John”
 - the value $v = \text{“John”}$ is accurate,
 - while the value $v_l = \text{“Jhn”}$ is inaccurate.

SYNTACTIC AND SEMANTIC ACCURACY

- Since usually we do not know, and it could be very costly to know, “the correct representation of the real world phenomenon that v aims to represent”, two kinds of accuracy can be identified, namely syntactic accuracy and semantic accuracy.
- *Syntactic accuracy* is defined when the vocabulary or reference domain of values is known (e.g. the domain of Italian first names, the domain of Czech municipalities), and is the degree to which values correctly represent the domain values of underlying vocabularies or reference domains, while
- *Semantic accuracy* is defined as the degree to which data values correctly represent the real world facts

EXAMPLE: CAN YOU FIGURE OUT WHICH VALUES ARE CORRECT?

1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prage	412776	null
2	Mrtin	Necasky	1982	7	8	Sq.	Wien	null	Bratislava	101278	Slovakia
3	Miroslav	Konecny	null	null	null	Str.	Saint Jon	49	Prague	412776	null
4	Calo	Btini	1949	June	7	Street	Dessiè	15	Rome	0019h	Italy
5	Miroslav	Knecy	1978	7	null	Sq.	Budapest	23	Wien	null	Austria
6	Anisa	Rula	1982	September	7	Street	Sesto	null	Mian	20...	Ital
7	Anita	Rula	1982	9	7	Via	Seto	23	Milan	null	Italy

NOW?

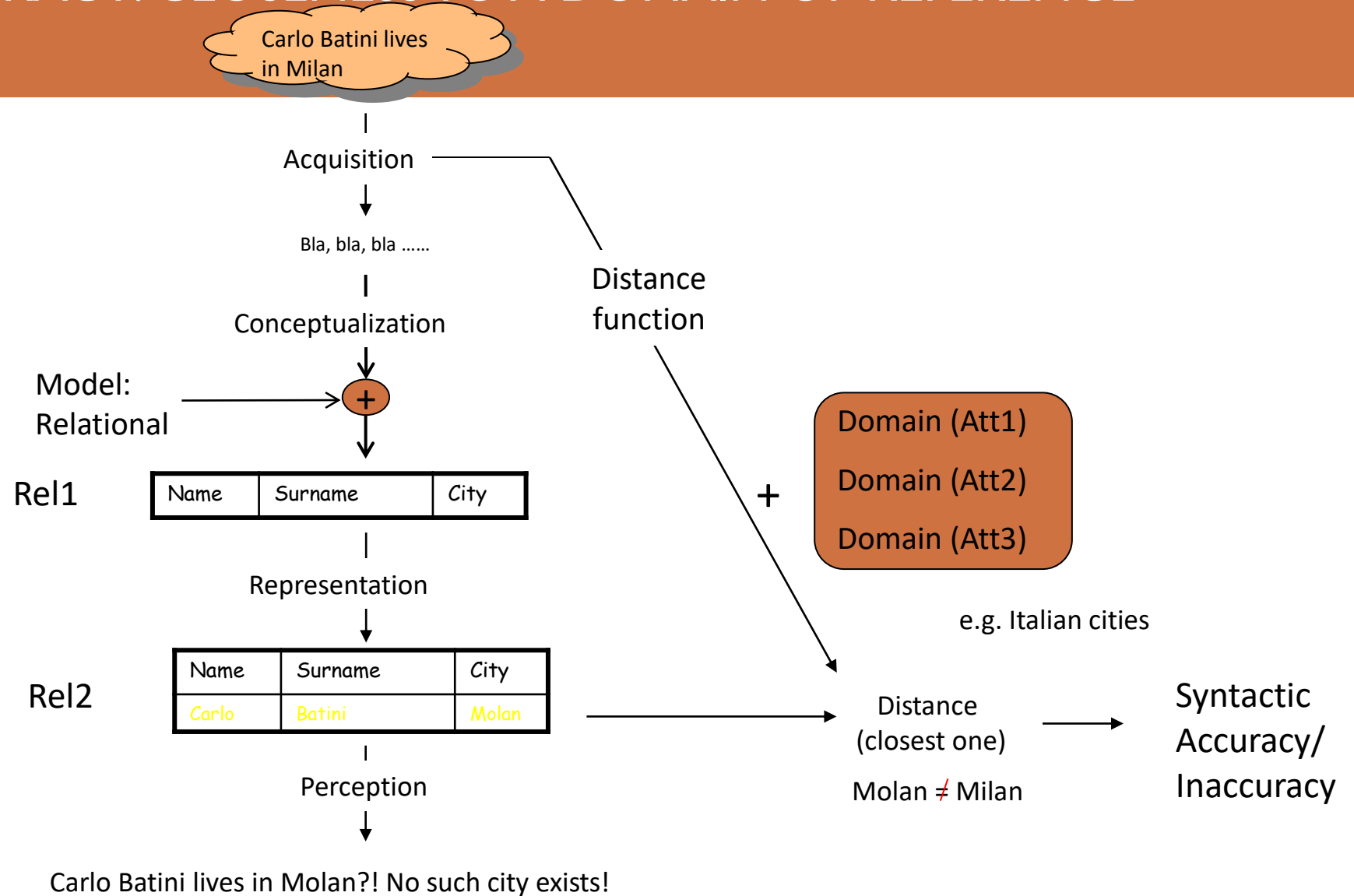
Tuple#	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name Toponym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prage	412776	null
2	Mrtin	Necasky	1982	7	8	Sq.	Wien	null	Bratislava	101278	Slovakia
3	Miroslav	Konecny	null	null	null	Str.	Saint Jon	49	Prague	412776	null
4	Calo	Btini	1949	June	7	Street	Dessiè	15	Rome	0019h	Italy
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6	Anisa	Rula	1982	September	7	Street	Sesto	null	Mian	20...	Ital
7	Anita	Rula	1982	9	7	Via	Seto	23	Milan	null	Italy



METRICS FOR SYNTACTIC ACCURACY



SYNTACTIC ACCURACY: CLOSENESS TO A DOMAIN OF REFERENCE



METRICS FOR SYNTACTIC ACCURACY IN STRUCTURED ALPHANUMERIC DATA

Two types

- Based on the alphanumeric strings
 - «Santa Margherita Ligure»
- Based on their structure in terms of items (tokens)
 - «Santa» «Margherita» «Ligure»

Metrics based on strings

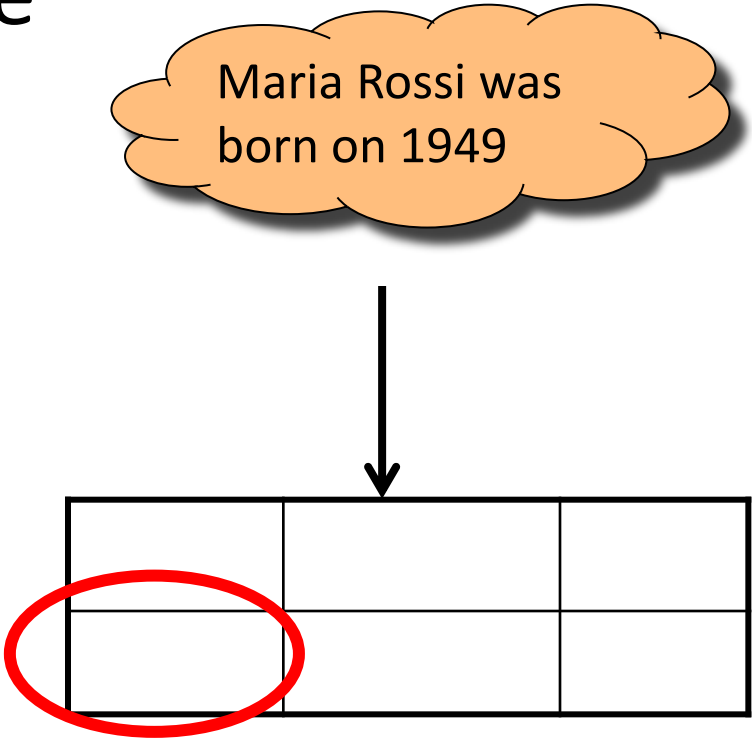
- Adopt a distance function, measured between the data value and the values of a reference domain, we will consider now:
- Edit distance
- Later on we will consider more distances

EDIT DISTANCE METRIC

- The unnormalized Edit Distance UED $[v1, v2]$ is equal to the number of insertions + deletions + substitutions of alphanumeric symbols necessary to transform $v1$ into $v2$.

UED(Mario, Carlo) = ??

Example



UED(Maro, Mario) = 1
UED(Maro, Maria) = 2
UED(Maro, Margherita) = 7
...

Nomi Italiani Maschili e Femminili - Lettera M

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Ring billigt med Comviq

Under hundraen med Fastpris Mini. Betala endast 95 kr/mån!



Quanti sono i nomi di battesimo italiani? Quasi impossibile dirlo ma sembra siano **oltre 10 mila**! Ovviamente molti di essi sono davvero più unici che rari. Ecco un elenco di circa 1700 nomi italiani (in **grassetto** i più comuni e diffusi), suddivisi tra maschili e femminili: alcuni potranno sembrare strani...pensa a quelli che non sono inclusi in questo elenco!

Nomi che iniziano con la lettera:

A B C D E F G I L M NO PQ R S TUV WZ

Maschili

Macario	Maccabeo
Maffeo	Maggiorino
Magno	Maiorico
Malco	Mamante
Mancio	Manetto
Manfredo	Manilio
Manlio	Mansueto
Manuele	Marcello
Marciano	Marco
Mariano	Marino
Mario	Marolo

Femminili

Macaria	Maddalena
Mafalda	Magda
Maida	Manuela
Mara	Marana
Marcella	Mareta
Margherita	Maria
Marianna	Marica
Mariella	Marilena
Marina	Marinella
Marinetta	Marisa
Marita	Marta

EDIT DISTANCE METRIC

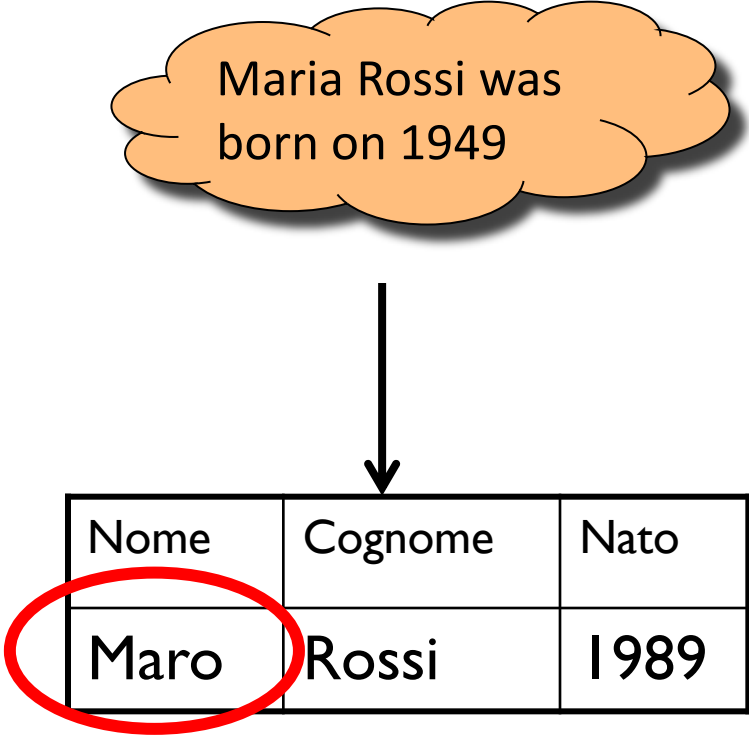
The unnormalized Edit Distance UED $[v1, v2]$ is equal to the number of insertions + deletions + substitutions of alphanumeric symbols necessary to transform $v1$ into $v2$.

The normalized Edit Distance $[v1, v2]$ where $v1$ and $v2$ are values in the domain D , where the maximum number of symbols is n , is

$$ED_{\text{norm}}(v1, v2) = 1 - ED(v1, v2)/n$$

- Values are within $[0, \dots, 1]$
- $ED_{\text{norm}}(\text{Mario}, \text{Carlo}) = (n = 10) \rightarrow ??$

Example



$n = 10$
 $Ed_n(\text{Maro}, \text{Mario}) = 1 - 1/10 = 0,9$
 $Ed_n(\text{Maro}, \text{Maria}) = 1 - 2/10 = 0,8$
 $Ed_n(\text{Maro}, \text{Margherita}) = 1 - 7/10 = 0,3$
...

Nomi Italiani Maschili e Femminili - Lettera M

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Google+

Twitter

Ring billigt med Comviq

Under hundraen med Fastpris Mini. Betala endast 95 kr/mån!

Quanti sono i nomi di battesimo italiani? Quasi impossibile dirlo ma sembra siano **oltre 10 mila**! Ovviamente molti di essi sono davvero più unici che rari. Ecco un elenco di circa 1700 nomi italiani (in **grassetto** i più comuni e diffusi), suddivisi tra maschili e femminili: alcuni potranno sembrare strani...pensa a quelli che non sono inclusi in questo elenco!

Nomi che iniziano con la lettera:
A B C D E F G I L M NO PQ R S TUV WZ

Maschili		Femminili	
Macario	Maccabeo	Macaria	Maddalena
Maffeo	Maggiorino	Mafalda	Magda
Magno	Maiorico	Maida	Manuela
Malco	Mamante	Mara	Marana
Mancio	Manetto	Marcella	Mareta
Manfredo	Manilio	Margherita	Maria
Manlio	Mansueto	Marianna	Marica
Manuele	Marcello	Mariella	Marilena
Marciano	Marco	Marina	Marinella
Mariano	Marino	Marinetta	Marisa
Mario	Marolo	Marita	Marta

EDIT DISTANCE METRICS

- The unnormalized Edit Distance UED $[v1, v2]$ is equal to the number of insertions + deletions + substitutions of alphanumeric symbols necessary to transform $v1$ into $v2$.
- The normalized Edit Distance $[v1, v2]$ where $v1$ and $v2$ are values in the domain D , where the maximum number of symbols is n , is

$$ED_{\text{norm}}(v1, v2) = 1 - ED(v1, v2)/n$$

- Values within $[0, \dots, 1]$
- The normalized Edit distance does not measure the distance but its complement to 1, i.e. the accuracy of $v1$ with respect to $v2$, considered a certified value:
- Accuracy = 1 corresponds to Distance = 0

COMPLETENESS

	Accuracy	Completeness	Currency	Consistency
Alphanumeric value		X		
Numerical value		X		
Tuple		X		
Attribute		X		
Relation		X		
Obejct		X		

COMPLETNESS

- **Completeness** (of a dataset), the coverage with which the observed phenomenon is represented in the dataset
- For example, for employees of an organization, a master data is complete when all employees are represented

Surname	Age	Place of birth
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

COMPLETENESS MEASURES FOR TABLES

1. Tuple completeness, presence of null values in a tuple
2. Attribute completeness, presence of null values in an attribute column
3. Table completeness, number of non null values in a table

Cognome	Eta'	Citta' Nascita
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

METRICS FOR COMPLETENESS

Attributes → Tuples	Attribute 1	Attribute 2		Attribute n
Tuple 1				
Tuple 2				
...				
Tuple m				

Type of completeness	Metrics
Tuple completeness (t_i)	# of null values in t_i / n
Attribute completeness (A_j)	# of null values in the column of A_j / m
Table completeness (T)	# of null values in T / $m * n$

Table T representing an universe U of k objects (e.g. Employees)

COMPLETENESS METRIC

Cognome	Eta'	Citta'Nascita
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

Attribute completeness

Cognome 100%

Eta' 33%

Citta'Nascita 66%

INSIGHTS INTO THE COMPLETENESS OF THE CLOSED OR OPEN WORLD ASSUMPTION

- In the previous definitions, we assumed a closed world assumption: all that is represented in the BD is true, all the rest is false. This is the typical assumption that is made in DBMSs.
- An alternative hypothesis is that of an open world: nothing is known about everything that is not represented.
- In this case we introduce the object completeness, which takes into account the fact that the objects that can be represented are more than the tuples of the table, and an indirect estimate is needed for this cardinality

COMPLETENESS

	accuracy	Completeness	currency	consistency
Alphanumeric value		X		
Numerical value		X		
tuple		X		
Relation		X		
Object		X		

COMPLETENESS MEASURES FOR TABLES

1. Tuple completeness, presence of null values in a tuple
2. Attribute completeness, presence of null values in an attribute column
3. Table completeness, number of non null values in a table
4. Object completeness, number of non null values in all the objects represented in tuples

Four objects



Cognome	Eta'	Citta' Nascita
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

COMPLETENES

Cognome	Eta'	Citta'Nascita
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

Employee

If there are 3 employees

Object completeness = 100%

Attribute completeness

Cognome 100%

Eta' 33%

Citta'Nascita 66%

COMPLETENES

Cognome	Eta'	Citta'Nascita
Rossi	null	null
Verdi	35	Roma
Neri	null	Milano

If there are 4 employees
Object completeness: 75 %

METRICS FOR COMPLETENESS

Attributes → Tuples	Attribute 1	Attribute 2		Attribute n
Tuple 1				
Tuple 2				
...				
Tuple m				

k
objects

Type of completeness	Metrics
Object completeness (T,U)	m/k
Tuple completeness (t_i)	# of null values in t_i / n
Attribute completeness (A_j)	# of null values in the column of A_j / m
Table completeness (T)	# of null values in T / $m * n$

Table T representing an universe U of k objects (e.g. Employees)

A MORE COMPLEX EXAMPLE

StudentID	Name	Surname	Vote	Date
6754	Mike	Collins	29	07/null/2004
8907	null	Herbert	18	07/17/2004
6578	Julianne	Merrals	null	07/17/2004
0987	Robert	Archer	null	null
1243	null	null	26	09/30/2004
2134	Bridget	Abbott	30	09/30/2004
6784	John	Miller	30	null
0098	Carl	Adams	25	09/30/2004
1111	John	null	28	09/30/2004
2564	Edward	Monroe	null	null
8976	Anthony	White	21	null
8973	Marianne	Collins	30	10/15/2004

TEMPORAL QUALITY DIMENSION

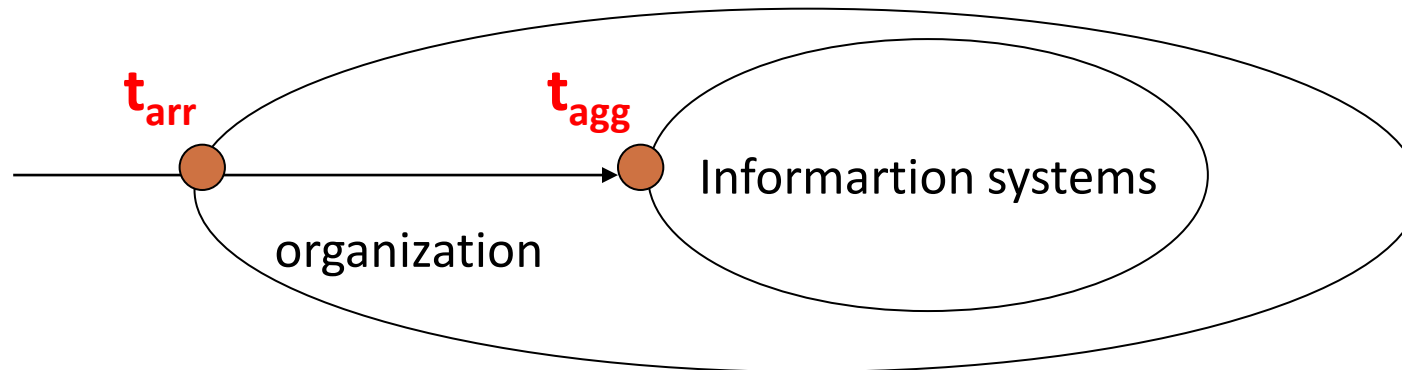
- Currency
- Timeliness

CURRENCY

- The currency measures how quickly the data is updated (compared to the corresponding phenomenon in the real world) E.g. the grade of an exam in the bd students-courses-exams.
- A first measure of the currency is the time delay between the time t_1 of the event in the real world that caused the variation of the data, and the instant t_2 of its registration in the information system
- Ex: Tax declarations are entered on average six months after submission → Currency = 6 months
- This measure is expensive because generally the event is not known. Eg change of address.

ANOTHER METRICS

- Currency as the difference between the time of arrival at the organization and the time in which the update is performed.
- Measurable if there is a log of arrivals and updates.



ANOTHER METRICS

- Difference from the "last updated" metadata
- For values with known update frequency, currency calculable in an approximate but inexpensive way.

TIMELINESS

- Timeliness measures how up-to-date the data is with respect to a particular process (or processes) using it.
- Example: the timetable of lessons and classrooms must be available before the start of the courses
- Timeliness, unlike currency, depends on the process, and is associated with the time point in which it must be available for the process that uses the data
- There may be data with high currency, but now obsolete for the process that uses them.
- If the timetable was produced outside the maximum time and was loaded immediately after being produced, it is current but not timely

CONSISTENCY

	Accuracy	Completeness	Currency	Consistency
Alphanumeric value				X
Numerical value				X
Tuple				X
Relation				X
Object				

CONSISTENCY

- Two meanings
- 1. Consistency of the data with the integrity constraints defined on the schema
- E.g. ZIP Code must be consistent with City
- 2. Consistency of the different representations of the same object of reality present in the database
- Eg The address must be represented with the same format in all the databases in which it is defined.

CONSISTENCY

- Any violation of an integrity constraint defined on the database
- Eg in a relationship
- Person (FISCAL CODE, Surname, Name, Gender)
- Example of integrity constraint:
- The gender must be consistent with the tenth character of the FISCAL CODE

CONSISTENCY

- Consistency constraints (or business rules) can include:
- a single attribute (attribute values must fall within the $\{1..10\}$ domain);
- more than one attribute (postal code and city must not conflict);
- Attributes in multiple relationships (e.g. referential integrity)
- Expressed in terms of probabilities:
- Example: age > 70 and hair color = black $\rightarrow$ probability = 0.001

QUALITY TRADEOFF

- Consistency and completeness in the relational model can be irreconcilable when one wants to respect referential integrity
- In the statistical/economic/financial domain (e.g. increase in GDP in the last three months) there is a tradeoff between timeliness and accuracy/competence, favoring the latter.
- On the web, in order to "arrive first" with news, in many cases timeliness is preferred over accuracy and completeness (e.g. weather forecast, agency dispatch)



QUALITY ASSESSMENT AND IMPROVEMENT



GOAL OF THE IMPROVEMENT PHASE

The goal of the improvement phase is to clean data in such a way to increase their quality w.r.t to values assessed in the previous phase. Since quality is multidimensional, specific activities are needed for the different qualities.

In the improvement phase we can adopt two general strategies, namely

- *data-driven* and
- *process-driven*

DATA-DRIVEN STRATEGIES

- Data-driven strategies improve the quality of data by directly modifying the value of data through the comparison with other data considered of good quality. For example, obsolete data values are updated by refreshing a database characterized by higher currency.

ATTIVITA' BASATE SUI PROCESSI

A DATABASE IS LIKE A LAKE

To Clean Up The Lake, One Must First
Eliminate The Sources Of Pollutant





A CASE STUDY



CASE STUDY BASED ON EUROPEAN PROFESSIONALS

Tuple #	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name T.oonym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prague	412776	null
2	Martin	Necasky	1975	7	8	Sq.	Vienna	null	Bratislava	101278	Slovensko
3	Miroslav	Konecny	null	null	null	Str.	Saint Jon	49	Prague	412776	null
4	Crlo	Btini	1949	June	7	Street	Dessìè	15	Roma	00198	Italia
5	Miroslav	Knecy	1978	7	10	Sq.	Budapest	23	Wien	null	Austria
6	Anisa	Rula	1982	September	7	Via	Sesto	null	Milano	20...	Ital
7	Anita	Rula	1982	9	7	Via	Seto	23	Milan	null	Italy
8	Anna	Rla	null	null	null	Via	Sarca	336	Milano	null	Italy
9	Carlo	Batini	1949	6	7	V.	Beato Angelico	23	Milan	20133	Italy
10	Carla	Botni	1949	June	7	Av.	Charles	null	Prague	412733	null
11	Marta	Necasky	1976	11	8	null	null	null	Bratislava	112234	Slov.



NORMALIZATION STEP



NORMALIZATION OF MONTH OF BIRTH AND COUNTRY

Tuple #	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name T.oponym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prague	412776	null
2	Martin	Necasky	1975	7	8	Sq.	Vienna	null	Bratislava	101278	Slovensko
3	Miroslav	Konecny	null	null	null	Str.	Saint Jon	49	Prague	412776	null
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6	Anisa	Rula	1982	September	7	Via	Sesto	null	Milano	20...	Ital
7	Anita	Rula	1982	9	7	Via	Seto	23	Milan	null	Italy
8	Anna	Rla	null	null	null	Via	Sarca	336	Milano	null	Italy
9	Carlo	Batini	1949	6	7	V.	Beato Angelico	23	Milan	20133	Italy
10	Carla	Botni	1949	June	7	Av.	Charles	null	Prague	412733	null
11	Marta	Necasky	1976	11	8	null	null	null	Bratislava	112234	Slov.

OUTPUT OF THE NORMALIZATION STEP

Tuple #	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name T.oponym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prage	4127 6	null
2	Martin	Necasky	1975	7	8	Square	Vienna	null	Bratislava	1012 8	Slovakia
3	Miroslav	Konecny	null	null	null	Street	Saint Jon	49	Prague	4127 6	null
4	Crlo	Btini	1949	6	7	Street	Dessiè	15	Roma	0019	Italia
5	Miroslav	Knecy	1978	7	10	Square	Budapest	23	Wien	null	Austria
6	Anisa	Rula	1982	9	7	Street	Sesto	null	Mian	20...	Ital
7	Anita	Rula	1982	9	7	Street	Seso	23	Milan	null	Italy
8	Anna	Rla	null	null	null	Street	Sarca	336	Milano	null	Italy
9	Carlo	Batini	1949	6	7	Street	Beato Angelico	23	Milano	2013	Italy
10	Carla	Botni	1949	6	7	Avenue	Charles	null	Prague	4127 3	null
11	Marta	Necasky	1976	11	8	null	null	null	Bratislava	112234	Slovakia



EXERCISE – EVALUATION OF METRICS FOR ACCURACY ASSESSMENT



ASSESSMENT AND IMPROVEMENT OF ACCURACY

All values that do not exactly match with values in reference domains have to be corrected. To do so, we have to choose suitable **reference tables**.

E.g. as to the value “Saint Jon”, we may use

- either the Google Refine service, or else we have to access
- the table of street names of Prague, made available from the reference database for addresses RUIAN – register of territorial identification, addresses and real estates

IMPROVEMENT OF ACCURACY

- Accuracy improvement is performed substituting the incorrect value with the one at lower distance in the reference table, where the distance can be evaluated after a previous inspection of values and their characteristics
- The result of the accuracy improvement is shown in Figure →, where we have corrected
 - in tuple I →

SYNTACTIC ACCURACY – TUPLE I

Tuple #	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name T.oponym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prage	412776	null
2	Martin	Necasky	1975	7	8	Square	Vienna	null	Bratislava	101278	Slovakia
3	Miroslav	Konecny	null	null	null	Street	Saint Jon	49	Prague	412776	null
4	Crlo	Btini	1949	6	7	Street	Dessiè	15	Roma	00198	Italia
5	Miroslav	Knecy	1978	7	10	Square	Budapest	23	Wien	null	Austria
6	Anisa	Rula	1982	9	7	Street	Sesto	null	Mian	20...	Ital
7	Anita	Rula	1982	9	7	Street	Seso	23	Milan	null	Italy
8	Anna	Rla	null	null	null	Street	Sarca	336	Milano	null	Italy
9	Carlo	Batini	1949	6	7	Street	Beato Angelico	23	Milano	20133	Italy
10	Carla	Botni	1949	6	7	Avenue	Charles	null	Prague	412733	null
11	Marta	Necasky	1976	11	8	null	null	null	Bratislava	112234	Slov.

SYNTACTIC ACCURACY – TUPLE 1

Tuple #	F.Name	Last Name	Y.Of Birth	M. Birth	D. Birth	Toponym	Name T.oponym	Number T.	City	Zip Code	Country
1	Miroslav	Konecny	1978	7	10	Street	St. John	49	Prage	412776	null
2	Martin	Necasky	1975	7	8	Square	Vienna	null	Bratislava	101278	Slovakia
3	Miroslav	Konecny	null	null	null	Street	Saint Jon	49	Prague	412776	null
4	Crlo	Btini	1949	6	7	Street	Dessiè	15	Roma	00198	Italia
5	Miroslav	Knecy	1978	7	10	Square	Budapest	23	Wien	null	Austria
6	Anisa	Rula	1982	9	7	Street	Sesto	null	Mian	20...	Ital
7	Anita	Rula	1982	9	7	Street	Seso	23	Milan	null	Italy
8	Anna	Rla	null	null	null	Street	Sarca	336	Milano	null	Italy
9	Carlo	Batini	1949	6	7	Street	Beato Angelico	23	Milano	20133	Italy
10	Carla	Botni	1949	6	7	Avenue	Charles	null	Prague	412733	null
11	Marta	Necasky	1976	11	8	null	null	null	Bratislava	112234	Slov.